



Penrith Selective High School

Mathematics Extension 1

Trial HSC 2019

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- No correction tape or white out to be used
- A reference sheet is provided with this paper
- In Questions 11–14, show relevant mathematical reasoning and/ or calculations

	Prelim	Series/ Induction	Calculus	Inverse Fns	Rates of Change Growth & decay	Prob.	Motion	Total
M/C	/4	/1	/2		/1	/1	/1	/10
Q11	/3	/3	/5			/2	/2	/15
Q12				/3	/5		/7	/15
Q13		/4		/8		/3		/15
Q14	/6			/4	/3		/2	/15
Total	/13	/8	/7	/15	/9	/6	/12	/70

Student Number: _____

Teacher's Name: _____

Section I (10 marks)

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Green multiple-choice answer sheet for Questions 1–10.

1. The polynomial $3x^3 + 5x^2 - 7x - 10$ has zeros α , β and γ .

What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?

- A. $\frac{7}{10}$
- B. $-\frac{7}{10}$
- C. $\frac{1}{2}$
- D. $-\frac{1}{2}$
2. Solve $6|x + 3| - 2|x + 1| = 0$.
- A. -4 or -2.5
- B. 4 or 2.5
- C. -4 or 2.5
- D. 4 or -2.5
3. For what value of c will the line $5x - 12y + c = 0$ be a tangent to the circle $x^2 + y^2 = 4$?
- A. 78
- B. 13
- C. 52
- D. 26

4. Using Simpson's Rule with three functional values evaluate $\int_0^1 \sin^{-1} x \, dx$

- A. $\frac{7\pi}{18}$
- B. $\frac{\pi}{9}$
- C. $\frac{7\pi}{6}$
- D. $\frac{7\pi}{36}$

5. The derivative of $\ln (\sin 2x)$ is:

- A. $2\cot 2x$
- B. $2\tan 2x$
- C. $\cot 2x$
- D. $\tan 2x$

6. What is the exact value of $\tan 105^\circ$?

- A. $2 - \sqrt{3}$
- B. $2 + \sqrt{3}$
- C. $-2 - \sqrt{3}$
- D. $-2 + \sqrt{3}$

7. Three men and three boys are seating at a round table.

What is the probability that they are alternating?

- A. $\frac{1}{30}$
- B. $\frac{3}{10}$
- C. $\frac{1}{5}$
- D. $\frac{1}{10}$

8. A particle moves so that at time t seconds, its displacement x metres from a fixed point O is given by $x = \pi + \sin(\frac{\pi}{3}t)$. When does the particle first return to its starting point?

- A. 1.5
- B. 3
- C. 4.5
- D. 6

9. Express 0.233 223 322 332..... as a fraction in its simplest form.

- A. $\frac{233}{999}$
- B. $\frac{233}{909}$
- C. $\frac{212}{909}$
- D. $\frac{232}{999}$

10. The rate of decay of a radioactive substance is proportional to the mass of the substance present at any time. If a quarter of the substance decomposed in 100 years, what percentage of the original amount remains after 500 years?

- A. 9.8%
- B. 8.8%
- C. 31.6%
- D. 23.7%

Section II – 60 marks

- Attempt Questions 11–14
 - Allow about 1 hour and 45 minutes for this section
 - Answer each question in the appropriate writing booklet.
 - Extra writing booklets are available.
 - In Questions 11–14, your responses should include relevant mathematical reasoning and/ or calculations.
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Question 11 (15 marks)

Marks

(a) Solve $\frac{2x+1}{2x-1} > 2$ **3**

(b) How many terms are there in the geometric series:

$$3^{7m} + 3^{5m} + 3^{3m} + \dots + 3^{(1-2k)m}?$$
 3

(c) From the digits 1, 2, 3, 4 and 5 a five digit number is formed.

What is the probability that it is divisible by 2? **2**

(d) A particle moves with a constant acceleration of -3m/s^2 .

If the velocity is -12m/s at two seconds, find the velocity when $t = 3$. **2**

(e) Find the primitive of the following functions:

(i) $1 + x^2$ **1**

(ii) $\frac{3x}{\sqrt{1+x^2}}$ **2**

(iii) $\frac{3x}{1+x^2}$ **2**

Question 12 (15 marks)

Start a new booklet

Marks

- (a) Water is flowing into an inverted cone of base radius 6cm and perpendicular height 14cm, at a rate of $21\text{cm}^3/\text{s}$.

(i) Show that $\frac{dr}{dt} = \frac{9}{\pi r^2}$ **3**

- (ii) Find the rate at which the surface area of the water is increasing when the depth of water is 8cm. **2**

- (b) Find the exact value of $\cos(\sin^{-1}\frac{3}{4} + \cos^{-1}\frac{2}{3})$ giving justifications. **3**

- (c) An object is thrown from the top of the library block 5m above the level ground. It is thrown at an angle of 30° to the horizontal with a velocity of 50m/s. (Take g as 10m/s^2)

- (i) Write down the 6 equations of motion. **2**

- (ii) Find the greatest height reached. **2**

- (iii) A person standing on the ground 225m away from the foot of the library. Will the person be hit by the object? Give reasons for your answer. **3**

Question 13 (15 marks)

Start a new booklet

Marks

- (a) Prove by Mathematical Induction that $5^{2n} - 2^{3n}$ is always divisible by 17 for $n \geq 1$. **4**
- (b) The area under the curve $y = \sin^{-1} x$ from $x = 0$ to $x = \frac{1}{2}$ bounded by the x-axis is rotated about the y-axis. Find the exact volume of the solid of revolution. **3**
- (c) Evaluate exactly $\int_0^{\frac{1}{4}} \frac{10dx}{\sqrt{1-16x^2}}$ **3**
- (d) An event has a probability of success of $\frac{1}{3}$ in a single trial. If n trials are conducted, find the least value of n for which the probability of obtaining exactly four successes exceeds triple the probability of obtaining exactly three successes. **3**
- (e) Show that the derivative of $y = \tan^{-1} \sqrt{x}$ is $\frac{1}{2\sqrt{x}(1+x)}$ **2**

Question 14 (15 marks)

Start a new booklet

Marks

- (a) Neatly graph $y = \sin^{-1}(\cos x)$ for $0 \leq x \leq 3\pi$ **4**
- (b) The tangent at $P(2ap, ap^2)$ to $x^2 = 4ay$ meets the x-axis at R .
The tangent at $Q(2aq, aq^2)$ meets the y-axis at S . Find the locus of the midpoint RS , given that $pq = -2$. (You may assume that the tangent is in the form $y = tx - at^2$). **3**

- (c) A particle is moving on a straight line such that $x = 12 - 2^{1-t}$ where x is in metres and t is the time in seconds. Describe the motion. 2

- (d) AOB is a quadrant of a circle of radius 30cm.

P is a particle that moves on the arc AB about O at a constant rate.

It moves from A to B in 15 minutes.

$$\angle AOP = \theta$$

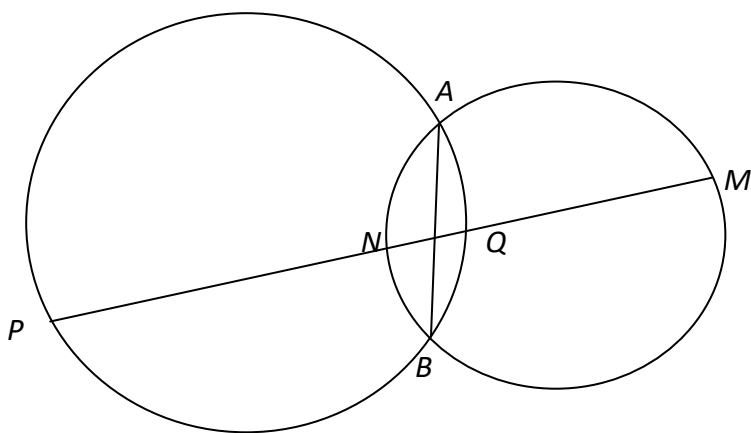
If Y is the total area of $\triangle OAP$ and $\triangle OBP$, find the rate at which Y is changing when $\theta = \frac{\pi}{6}$. 3

- (e) AB is the common chord. 3

$PNXQM$ is a straight line.

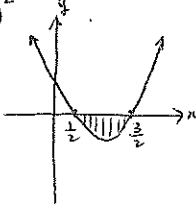
X is the point of intersection of AB with $PNQM$.

Copy the diagram into your booklet. Prove that $\frac{PN}{NX} = \frac{MQ}{QX}$ Giving reasons.



END OF EXAMINATION PAPER

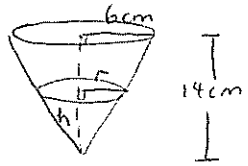
M/C 1.B 2.A 3.0 4.0 5.A 6.C 7.0 8.B 9.C 10.D

QUESTION: //	Markers Comments
11 a) $\frac{2x+1}{2x-1} > 2 \quad x \neq \frac{1}{2}$ Multiply both sides by $(2x-1)^2$ $(2x+1)(2x-1) > 2(2x-1)^2$ $4x^2 - 1 > 8x^2 - 8x + 2$ $4x^2 - 8x + 3 < 0$ ① $(2x-3)(2x-1) < 0$ $\therefore \frac{1}{2} < x < \frac{3}{2}$ ②  b) $a = 3^{7m}$, $r = 3^{-2m}$, $n = ?$, $T_n = 3^{(1-2k)m}$, G.P. ① $T_n = ar^{n-1}$ $3^{(1-2k)m} = 3^{7m} \cdot (3^{-2m})^{n-1}$ $3^{(1-2k)m} = 3^{7m-2m(n-1)}$ $(1-2k)m = 7m - 2m(n-1) \quad ①$ $2m(n-1) = 7m - m + 2km$ $n-1 = 3+k$ $\therefore n = 4+k \text{ terms} \quad ①$ c) Ending with 2 = 4! Ending with 4 = 4! $P(\text{divisible by 2}) = \frac{4! + 4!}{5!} \quad ①$ $= \frac{2}{5} \quad ①$ d) $\ddot{x} = -3$ $\dot{x} = -3t + C$ $t=2 \quad \dot{x} = -12 \quad \left. \begin{array}{l} \dot{x} = -3t + C \\ \dot{x} = -12 \end{array} \right\} C = -6 \quad ①$ $\dot{x} = -3t - 6$ Sub $t=3$, $\dot{x} = -9-6$ $= -15 \text{ m/s} \quad ①$	$\frac{2x+1}{2x-1} > 2$ $2x+1 > 4x-2$ $3 > 2x$ $x < \frac{3}{2} \quad \text{1 mark only}$ * Alternative $a = 7m, d = -2m, T_n = (1-2k)m, \text{ A.P.}$ $T_n = a + (n-1)d$ $(1-2k)m = 7m + (n-1)(-2m)$ $1-2k = 7-2(n-1)$ $1-2k = 9-2n$ $2n = 2k+8$ $n = k+4$

QUESTION: //	Markers Comments
11 e) i) $\int (1+x^2) dx = x + \frac{x^3}{3} + C \quad ①$ ii) $\int \frac{3x}{\sqrt{1+x^2}} dx$ Let $u = 1+x^2$ $= \frac{3}{2} \int \frac{1}{\sqrt{u}} du \quad ① \quad \frac{du}{dx} = 2x$ $= \frac{3}{2} (2u^{\frac{1}{2}}) + C \quad \frac{1}{2} du = x dx$ $= 3\sqrt{1+x^2} + C \quad ①$ iii) $\int \frac{3x}{1+x^2} dx$ $= \frac{3}{2} \int \frac{2x}{1+x^2} dx \quad ①$ $= \frac{3}{2} \log_e (1+x^2) + C \quad ①$	* Some students got mixed up with $\sin^{-1}x$ * Some students got mixed with $\tan^{-1}x$.

Q12

a) i)



$$\frac{r}{h} = \frac{6}{14}$$

$$6h = 14r$$

$$h = \frac{7r}{3} \quad \checkmark \quad \textcircled{1}$$

Given $\frac{dV}{dt} = 21$

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$21 = \frac{7\pi}{3} r^2 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = 21 \times \frac{3}{7\pi r^2} \quad \checkmark \quad \textcircled{1}$$

$$= \frac{9}{\pi r^2}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{7r}{3} \times \pi r^2$$

$$V = \frac{7\pi}{9} r^3$$

$$\frac{dV}{dr} = \frac{7\pi}{3} r^2 \quad \checkmark \quad \textcircled{1}$$

* This question was very poorly attempted.

Using similar triangles to find $h = \frac{7r}{3}$

ii) $A = \pi r^2$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times \frac{9}{\pi r^2}$$

$$= \frac{18}{r}$$

when $h=8$,

$$8 = \frac{7r}{3}$$

$$r = 8 \times \frac{3}{7}$$

$$r = \frac{24}{7} \quad \checkmark \quad \textcircled{1}$$

$$\frac{dA}{dt} = 18 \times \frac{7}{24}$$

$$= \frac{21}{4}$$

$\therefore$ rising at $\frac{21}{4} \text{ cm}^2/\text{s} \quad \checkmark \quad \textcircled{1}$

* poorly attempted

* Most students failed to find the correct 'r'.

Q12

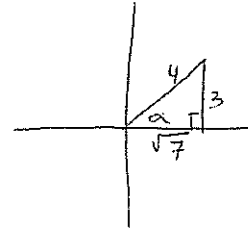
b) i) Let $\alpha = \sin^{-1} \frac{3}{4}$

$$\sin \alpha = \frac{3}{4}$$

$$-\frac{\pi}{2} \leq \alpha \leq \frac{\pi}{2}$$

but $\sin \alpha > 0$

$$\therefore 0 < \alpha < \frac{\pi}{2}$$



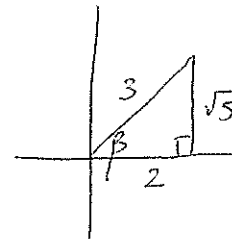
Let $\beta = \cos^{-1} \frac{2}{3}$

$$\cos \beta = \frac{2}{3}$$

$$0 \leq \beta \leq \pi$$

but $\cos \beta > 0$

$$\therefore 0 < \beta < \frac{\pi}{2} \quad \checkmark \quad \textcircled{1}$$



* Need to justify why α and β is in first quadrant to get 1 mark

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \frac{\sqrt{7}}{4} \times \frac{2}{3} - \frac{3}{4} \times \frac{\sqrt{5}}{3} \quad \checkmark \quad \textcircled{1}$$

$$= \frac{\sqrt{7}}{6} - \frac{\sqrt{5}}{4}$$

$$= \frac{2\sqrt{7} - 3\sqrt{5}}{12} \quad \checkmark \quad \textcircled{1}$$

* Most students found the exact value but did not justify why α and β was in first quadrant, 1 mark was penalised.

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Q12

$$\begin{aligned} \text{c) (i)} \quad \ddot{x} &= 0 \\ \dot{x} &= 50 \cos 30^\circ = 25\sqrt{3} \\ x &= 25\sqrt{3}t \end{aligned} \quad \left. \vphantom{\begin{aligned} \ddot{x} &= 0 \\ \dot{x} &= 50 \cos 30^\circ = 25\sqrt{3} \\ x &= 25\sqrt{3}t \end{aligned}} \right\} \checkmark \textcircled{1}$$

$$\begin{aligned} \ddot{y} &= -10 \\ \dot{y} &= -10t + 25 \\ y &= -5t^2 + 25t + 5 \end{aligned} \quad \left. \vphantom{\begin{aligned} \ddot{y} &= -10 \\ \dot{y} &= -10t + 25 \\ y &= -5t^2 + 25t + 5 \end{aligned}} \right\} \checkmark \textcircled{1}$$

* Working out not necessary for this question.

ii) Max height when $\dot{y} = 0$

$$\begin{aligned} -10t + 25 &= 0 \\ 10t &= 25 \\ t &= 2.5 \quad \checkmark \textcircled{1} \end{aligned}$$

$$\begin{aligned} y &= -5(2.5)^2 + 25(2.5) + 5 \\ y &= 36.25 \text{ m} \quad \checkmark \textcircled{1} \end{aligned}$$

* Well done.

* Carried forward marks was awarded if t was incorrect or equation 'y' was wrong in part (i).

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Q12.

$$\begin{aligned} \text{c) iii)} \quad \text{Let } y &= 0 \\ -5t^2 + 25t + 5 &= 0 \\ t^2 - 5t - 1 &= 0 \end{aligned}$$

$$\begin{aligned} t &= \frac{5 \pm \sqrt{25 - (4)(-1)}}{2} \\ &= \frac{5 \pm \sqrt{29}}{2} \quad \checkmark \textcircled{1} \end{aligned}$$

but $t \geq 0$

$$\begin{aligned} x &= 25\sqrt{3}t \\ &= 25\sqrt{3} \times \frac{5 + \sqrt{29}}{2} \\ &= 224.85 \text{ m} \quad \checkmark \textcircled{1} \end{aligned}$$

∴ The person will not be hit by the object since $224.85 < 225 \text{ m}$ $\checkmark \textcircled{1}$

* common error : quadratic equation

$$t = \frac{5 \pm \sqrt{24}}{2}$$

CfE marks was awarded if t was wrong.

* part (c) was the best attempted in Q12.

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Q13

a) Step 1: Prove true for $n=1$

$$5^2 - 2^3 = 17$$

$\therefore$ True for $n=1$ ✓

①

Step 2: Assume true for $n=k$

$$5^{2k} - 2^{3k} = 17P \quad \text{where } P \text{ is an integer}$$

$$\therefore 5^{2k} = 17P + 2^{3k}$$

Must write

✓ ①

Step 3: Prove true for $n=k+1$

$$5^{2(k+1)} - 2^{3(k+1)} = 17Q \quad \text{where } Q \text{ is integer}$$

$$\text{LHS} = 5^{2(k+1)} - 2^{3(k+1)}$$

$$= 5^2(5^{2k}) - 2^3(2^{3k})$$

$$= 25(17P + 2^{3k}) - 8(2^{3k})$$

By assumption ✓ ①

$$= 425P + 17(2^{3k})$$

$$= 17(25P + 2^{3k}) \quad \checkmark \text{ ①}$$

$$= 17Q$$

$$= \text{RHS}$$

Step 4: By the principle of mathematical induction

It is true for all integer $n \geq 1$

* This question was very well done, marks are penalised if students didn't state 'P is an integer'!

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Q13

b) $y = \sin^{-1} x$ when $x=0$ $y=0$

$$x = \sin y$$

$$x = \frac{1}{2} \quad y = \frac{\pi}{6}$$

$$V = \pi \int_0^{\frac{\pi}{6}} \sin^2 y \, dy$$

$$V = \pi r^2 h = \pi \left(\frac{1}{2}\right)^2 \frac{\pi}{6}$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{6}} (1 - \cos 2y) \, dy = \frac{\pi^2}{24}$$

$$= \frac{\pi}{2} \left[y - \frac{1}{2} \sin 2y \right]_0^{\frac{\pi}{6}} \quad \text{①}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{6} - \frac{1}{2} \sin \frac{\pi}{3} \right) - \left(0 - \frac{1}{2} \sin 0 \right) \right]$$

$$= \frac{\pi}{2} \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) \quad \text{①}$$

$$= \frac{\pi(2\pi - 3\sqrt{3})}{24} \text{ units}^3 \quad \text{①}$$

* Badly done $\therefore V = \frac{\pi^2}{24} - \frac{2\pi^2}{24} + \frac{3\sqrt{3}}{24} = \frac{3\sqrt{3}\pi - \pi^2}{24}$

* common errors, forgot to change $x = \frac{1}{2}$ to $y = \frac{\pi}{6}$

* integrated $1 - \cos 2y$ incorrectly

* incorrect substitution

* most students only found volume of the area bounded by the y-axis...

Pennith Selective Trial HSC Ex1 2019.

Q13

$$\begin{aligned}
 c) \int_0^{\frac{1}{4}} \frac{10}{\sqrt{1-16x^2}} dx &= \frac{10}{4} \int_0^{\frac{1}{4}} \frac{1}{\sqrt{\frac{1}{16}-x^2}} \\
 &= \frac{10}{4} \left[\sin^{-1} \left(\frac{x}{\frac{1}{4}} \right) \right]_0^{\frac{1}{4}} \\
 &= \frac{10}{4} \left[\sin^{-1}(4x) \right]_0^{\frac{1}{4}} \checkmark \textcircled{1} \\
 &= \frac{5}{2} (\sin^{-1} 1 - \sin^{-1} 0) \checkmark \textcircled{1} \\
 &= \frac{5}{2} \left(\frac{\pi}{2} - 0 \right) \\
 &= \frac{5\pi}{4} \checkmark \textcircled{1} \quad \text{Well done!}
 \end{aligned}$$

$$\begin{aligned}
 d) P(r=3) &= {}^nC_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{n-3} \\
 P(r=4) &= {}^nC_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{n-4} \\
 {}^nC_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^{n-4} &> {}^nC_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{n-3} \times 3 \checkmark \textcircled{1} \\
 \frac{n!}{4!(n-4)!} \times \frac{1}{3} &> 3 \times \frac{n!}{3!(n-3)!} \times \frac{2}{3} \\
 \frac{(n-3)!}{(n-4)!} &> 24 \checkmark \textcircled{1} \\
 n-3 &> 24 \\
 n &> 27 \\
 \text{smallest value } n &= 28 \checkmark \textcircled{1}
 \end{aligned}$$

Pennith Selective Ex1 HSC Trial 2019.

Q13

$$\begin{aligned}
 e) y &= \tan^{-1} \sqrt{x} \\
 y &= \tan^{-1} u \quad \text{Let } u = \sqrt{x} \\
 \frac{du}{dx} &= \frac{1}{2\sqrt{x}} \checkmark \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \\
 &= \frac{1}{1+u^2} \times \frac{1}{2\sqrt{x}} \\
 &= \frac{1}{1+(\sqrt{x})^2} \times \frac{1}{2\sqrt{x}} \checkmark \textcircled{1} \\
 &= \frac{1}{2\sqrt{x}(1+x)}
 \end{aligned}$$

* This is a show question, hence you must show every step of working out, no matter how trivial it is.

* Marks was penalised if the answer $\frac{1}{2\sqrt{x}(1+x)}$ just magically appeared in the final step without any justification.

Examination:

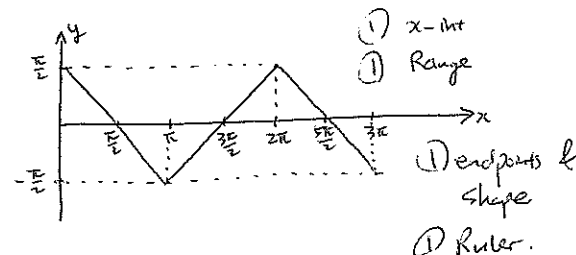
Level:

Year:

QUESTION: 14

$$14 a) y = \sin^{-1}(\cos x) \\ = \sin^{-1}[\sin(\frac{\pi}{2} - x)] \\ = \frac{\pi}{2} - x$$

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, \quad x = \text{all real numbers.}$$



b) Equation of tangent at P, $y = px - ap^2$
 Sub $y=0$ for R,
 $0 = px - ap^2$
 $x = ap$ $\therefore R(ap, 0)$

Equation of tangent at Q, $y = qx - aq^2$
 Sub $x=0$ for S,
 $y = -aq^2$ $\therefore S(0, -aq^2)$ ①

$$\therefore M\left(\frac{ap}{2}, -\frac{aq^2}{2}\right) \quad ①$$

$$x = \frac{ap}{2}, \quad y = -\frac{aq^2}{2}$$

$$p = \frac{2x}{a}, \quad q^2 = -\frac{2y}{a}$$

$$p^2 = \frac{4x^2}{a^2}$$

$$pq = -2$$

$$p^2 q^2 = 4$$

$$\frac{4x^2}{a^2} \times \left(-\frac{2y}{a}\right) = 4$$

$$-8x^2 y = 4a^3$$

$$y = -\frac{a^3}{2x^2} \quad ①$$

Markers Comments

* Label the axes

* Scales on the x-axis need to be equally spaced

* Some students drew a curve instead of a straight line.

* Don't need to derive the equation of the tangent by differentiating the curve, it was given in the question. Just replace the parameter t by p and q

* Most students worked out the coordinates of the point R and P and the midpoint, but some struggled to get the cartesian equation of the locus by eliminating p and q

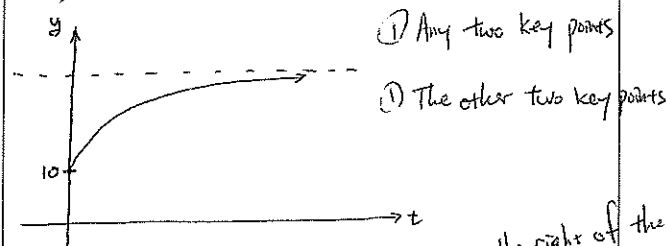
Examination:

Level:

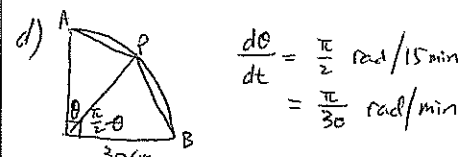
Year:

QUESTION:

$$14 c) x = 12 - 2^{1-t}$$



The particle was initially at $x=10$, moving to the right. It slows down and approaches to $x=12$, but never actually reaches it or stops



$$Y = \frac{1}{2}(30)^2 \sin \theta + \frac{1}{2}(30)^2 \sin\left(\frac{\pi}{2} - \theta\right) \\ = 450 \sin \theta + 450 \cos \theta \quad ①$$

$$\frac{dY}{d\theta} = 450 \cos \theta - 450 \sin \theta \quad ①$$

$$\frac{dY}{dt} = \frac{dY}{d\theta} \times \frac{d\theta}{dt}$$

$$= 450(\cos \theta - \sin \theta) \times \frac{\pi}{30} \quad ①$$

$$\text{When } \theta = \frac{\pi}{6}, \quad \frac{dY}{dt} = \frac{450\pi}{30} \left(\cos \frac{\pi}{6} - \sin \frac{\pi}{6}\right)$$

$$= 15\pi \left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)$$

$$= \frac{15\pi}{2} (\sqrt{3} - 1) \text{ cm/min} \quad ①$$

Markers Comments

* Poorly done

$$\neq \frac{d\theta}{dt} \neq \frac{360^\circ}{15 \text{ min}} \text{ or } 6^\circ/\text{min}$$

The angle needs to be in radians

* Many students found the area of sectors using $\frac{1}{2}r^2\theta$ instead of the two triangles
 $Y = \text{total area of the two } \Delta's$

* 1 mark was given if you have at least attempted to apply the chain rule in terms of $\frac{dY}{d\theta}$ and $\frac{d\theta}{dt}$

Examination:

Level:

Year:

QUESTION: 14	Markers Comments
14 e) $(PX)(XQ) = (AX)(XB)$ (Products of intercepts of chords in a circle) Similarly $(MX)(XN) = (AX)(XB)$ (1) $\therefore (PX)(XQ) = (MX)(XN)$ $(PN + NX)(XQ) = (MQ + QX)(XN)$ $(PN)(XQ) + (NX)(XQ) = (MQ)(XN) + (QX)(XN)$ (1) $(PN)(XQ) = (MQ)(XN)$ Divide both sides by $(XN)(XQ)$ (1) $\therefore \frac{PN}{NX} = \frac{MQ}{QX}$	